

Surds

Keep the root exact. Split out square factors, expand like brackets, and clear a surd from the bottom with its conjugate.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Surds

01 Simplify a surd

$$\sqrt{20} = 2\sqrt{5} \quad \sqrt{45} + \sqrt{20} \quad \sqrt{675} = n\sqrt{27} \quad (\sqrt{3})^5$$

02 Expand brackets with surds

$$(\sqrt{12} - \sqrt{3})^2 \quad (8 - \sqrt{x})(5 + \sqrt{x}) \quad (a - b)(a + b)$$

03 Rationalise: one term

$$10/\sqrt{5} \quad (\sqrt{20} + \sqrt{80})/\sqrt{3} \quad \text{spot the mistake}$$

04 Rationalise: two terms

$$6/(3 - \sqrt{7}) \quad \text{"in the form } a + \sqrt{b}\text{"} \quad (\sqrt{2} - 1)^2 \text{ on the bottom}$$

05 Surds in problems

$$\text{rectangle width} \quad \text{cuboid volume} \quad \text{maximum area}$$

THREE RULES Everything in this topic comes from these:

ZONE 1

$$\sqrt{ab} = \sqrt{a}\sqrt{b}$$

split out a square factor:

$$\sqrt{12} = \sqrt{4}\sqrt{3} = 2\sqrt{3}$$

ZONE 2

$$\sqrt{a} \times \sqrt{a} = a$$

a surd times itself is a whole number

ZONE 3

$$(a + \sqrt{b})(a - \sqrt{b})$$

= $a^2 - b$: the conjugate kills the surd

01 Simplify a Surd

Find the biggest square number that divides the number under the root. Take its root outside. Only **like** surds add.

WHEN YOU SEE → YOU DO

“without a calculator, show that $\sqrt{20} = 2\sqrt{5}$ ”

→ $\sqrt{4 \times 5} = \sqrt{4}\sqrt{5}$

$\sqrt{45} + \sqrt{20}$

→ simplify each first: $3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$

“in the form $n\sqrt{27}$ ”

→ divide: $\sqrt{675} \div \sqrt{27} = \sqrt{25} = 5$

$(\sqrt{3})^5$

→ pairs: $\sqrt{3}\sqrt{3}\sqrt{3}\sqrt{3}\sqrt{3} = 9\sqrt{3}$

Biggest square factor

Split the root

Root the square

square numbers	4	9	16	25	36	49	64	100
example	$\sqrt{12}$	$\sqrt{45}$	$\sqrt{32}$	$\sqrt{75}$	$\sqrt{72}$	$\sqrt{98}$	$\sqrt{128}$	$\sqrt{300}$
simplified	$2\sqrt{3}$	$3\sqrt{5}$	$4\sqrt{2}$	$5\sqrt{3}$	$6\sqrt{2}$	$7\sqrt{2}$	$8\sqrt{2}$	$10\sqrt{3}$

× **TRAP** $\sqrt{9} + \sqrt{16} \neq \sqrt{25}$. Roots do not add like that: $3 + 4 = 7$, not 5.

IDEA 01

Practise: Simplify a Surd

My questions first, then selected practice.

Q1 **DR ESLAM** Square factor

CH 1 • TOPIC 10 2 marks

Simplify (a) $\sqrt{50}$ (b) $\sqrt{12} + \sqrt{27}$ **SELECTED PRACTICE****Q2** **PRACTICE** Practice • Medium

CH 1 • TOPIC 10 2 marks

3 Show that $\frac{10}{\sqrt{5}}$ can be rewritten as $2\sqrt{5}$

(2 marks)

Q3**PRACTICE** Practice • Medium

CH 1 • TOPIC 10 2 marks

- 7 Show that $\sqrt{45} + \sqrt{20} = 5\sqrt{5}$
Show your working clearly.

(2 marks)

Q4**PRACTICE** Practice • Medium

CH 1 • TOPIC 10 2 marks

- 8 Without using a calculator, show that $\sqrt{20} = 2\sqrt{5}$.

(2 marks)

Q5**PRACTICE**

Practice · Hard

CH 1 · TOPIC 10

3 marks

- 5 Show that $\sqrt{5}(\sqrt{8} + \sqrt{18})$ can be written in the form $a\sqrt{10}$ and state the value of a . You must show your full working.

(3 marks)**Q6****PRACTICE**

Practice · Hard

CH 1 · TOPIC 10

5 marks

- 7 (a) Show that $\sqrt{3} + \sqrt{12}$ can be rewritten as $3\sqrt{3}$

(2 marks)

(b) Show that $\left(\frac{1}{\sqrt{3}}\right)^7 = \frac{\sqrt{3}}{81}$

(3 marks)

02 Expand Brackets with Surds

Expand exactly like algebra (grid or FOIL). Every $\sqrt{a} \times \sqrt{a}$ becomes a ; collect whole numbers and surds separately.

WHEN YOU SEE → YOU DO

$$(\sqrt{12} - \sqrt{3})^2$$

→ write it twice: $(\)(\)$ — not $12 - 3$

$$(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$$

→ expand, then match the whole numbers and the surds

$$(a - b)(a + b) \text{ with } a = \sqrt{8} + 4$$

→ use $a^2 - b^2$, or subtract then add first

“simplifies to an integer”

→ show the surd terms cancel

WORKED MODEL • JANUARY 2021 P1H

$(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$, x prime, y integer. Find x and y .

1 $40 + 8\sqrt{x} - 5\sqrt{x} - x$

expand

2 $(40 - x) + 3\sqrt{x}$

collect

3 $40 - x = 21, 3 = y$

match the parts

ANSWER $x = 19, y = 3$

IDEA 02

Practise: Expand Brackets with Surds

My questions first, then selected practice, then real past papers.

Q7 DR ESLAM Expand

CH 1 • TOPIC 10 2 marks

Expand and simplify $(2 + \sqrt{3})^2$.

SELECTED PRACTICE**Q8** PRACTICE Practice • Medium

CH 1 • TOPIC 10 4 marks

1 (a) Show that $\frac{12}{\sqrt{3}}$ can be rewritten as $4\sqrt{3}$

(2 marks)

(b) Show that $(\sqrt{2} + \sqrt{8})^2 = 18$

(2 marks)

Q9**PRACTICE**

Practice • Hard

CH 1 • TOPIC 10 4 marks

1 (a) Show that $\frac{5}{\sqrt{2}}$ can be written as $\frac{5\sqrt{2}}{2}$

(2 marks)

(b) Show that $(2 + \sqrt{3})^2 - (2 - \sqrt{3})^2 = 8\sqrt{3}$

(2 marks)

Q10**PRACTICE**

Practice • Hard

CH 1 • TOPIC 10 3 marks

4 Simplify fully $\frac{(6 - \sqrt{5})(6 + \sqrt{5})}{\sqrt{31}}$

You must show your working.

(3 marks)

Q11**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 10 4 marks

$$1 \quad a = \sqrt{8} + 2$$

$$b = \sqrt{8} - 2$$

$$T = a^2 - b^2$$

Show that $T = c\sqrt{2}$ stating the value of c

(4 marks)

PAST PAPERS**Q12****PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q18

CH 1 • TOPIC 10 3 marks

- 18 Given that $(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$ where x is a prime number and y is an integer, find the value of x and the value of y . Show each stage of your working clearly.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$

(Total for Question 18 is 3 marks)

Q13**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q13

CH 1 • TOPIC 10**3 marks**

13 $a = \sqrt{8} + 4$

$b = \sqrt{8} - 4$

 $(a - b)(a + b)$ can be written in the form $y\sqrt{4y}$ Find the value of y

Show your working clearly.

$y = \dots\dots\dots$

(Total for Question 13 is 3 marks)

03 Rationalise: One Term

A single surd on the bottom? Multiply top and bottom by that surd. The bottom becomes a whole number.

WHEN YOU SEE → YOU DO

$$\frac{10}{\sqrt{5}}$$

$$\rightarrow \times \frac{\sqrt{5}}{\sqrt{5}}: \frac{10\sqrt{5}}{5} = 2\sqrt{5}$$

$$\frac{5}{\sqrt{12}}$$

$$\rightarrow \text{simplify first: } \sqrt{12} = 2\sqrt{3}, \text{ then } \times \sqrt{3}$$

“find her mistake”

$$\rightarrow \text{check each line: } \sqrt{12} \times \sqrt{12} = 12, \text{ and } \sqrt{12} = 2\sqrt{3}$$

$$\frac{\sqrt{20} + \sqrt{80}}{\sqrt{3}} \quad \text{“in the form } \sqrt{a}\text{”}$$

$$\rightarrow \text{simplify the top, then square it all}$$

WORKED MODEL • ONE SURD

Rationalise $\frac{5}{\sqrt{12}}$.

$$1 \quad \frac{5}{2\sqrt{3}}$$

simplify first

$$2 \quad \frac{5 \times \sqrt{3}}{2\sqrt{3} \times \sqrt{3}} = \frac{5\sqrt{3}}{6}$$

$\times \sqrt{3}$ top and bottom

ANSWER $\frac{5\sqrt{3}}{6}$

IDEA 03

Practise: Rationalise: One Term

My questions first, then selected practice.

Q14 DR ESLAM One term

CH 1 • TOPIC 10 2 marks

Rationalise the denominator of $\frac{6}{\sqrt{2}}$.

SELECTED PRACTICE**Q15** PRACTICE Practice • Medium

CH 1 • TOPIC 10 3 marks

6 Show that $\frac{\sqrt{20} + \sqrt{80}}{\sqrt{3}}$ can be expressed in the form $\sqrt{a}$ where a is an integer.

Show your working clearly.

(3 marks)

Q16

PRACTICE

Practice • Hard

CH 1 • TOPIC 10 2 marks

3 (a) Martin did this question.

Rationalise the denominator of $\frac{14}{2 + \sqrt{3}}$

Here is how he answered the question.

$$\begin{aligned}\frac{14}{2 + \sqrt{3}} &= \frac{14 \times (2 - \sqrt{3})}{(2 + \sqrt{3})(2 - \sqrt{3})} \\ &= \frac{28 - 14\sqrt{3}}{4 + 2\sqrt{3} - 2\sqrt{3} + 3} \\ &= \frac{28 - 14\sqrt{3}}{7} \\ &= 4 - 2\sqrt{3}\end{aligned}$$

Martin's answer is wrong.

Find Martin's mistake.

(1 mark)

(b) Sian did this question.

Rationalise the denominator of $\frac{5}{\sqrt{12}}$

Here is how she answered the question.

$$\begin{aligned}\frac{5}{\sqrt{12}} &= \frac{5\sqrt{12}}{\sqrt{12} \times \sqrt{12}} \\ &= \frac{5 \times 3\sqrt{2}}{12} \\ &= \frac{5\sqrt{2}}{4}\end{aligned}$$

Sian's answer is wrong.

Find Sian's mistake.

(1 mark)

Q17**PRACTICE**

Practice • Hard

CH 1 • TOPIC 10

3 marks

- 15 Express $\frac{\sqrt{125} + \sqrt{80}}{\sqrt{3}}$ in the form $\sqrt{n}$ where n is an integer.

Show your working clearly.

(3 marks)**Q18****PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 10

3 marks

- 3 Show that $\frac{1}{1 + \frac{1}{\sqrt{2}}}$ can be written as $2 - \sqrt{2}$

(3 marks)

04 Rationalise: Two Terms — the Conjugate

Bottom is $a + \sqrt{b}$? Multiply top and bottom by $a - \sqrt{b}$ (change the sign). The bottom becomes $a^2 - b$.

WHEN YOU SEE → YOU DO

$$\frac{6}{3 - \sqrt{7}}$$

→ $\times \frac{3 + \sqrt{7}}{3 + \sqrt{7}}$, bottom = $9 - 7 = 2$

“in the form $a + \sqrt{b}$ / $a - \sqrt{b}$ ”

→ rationalise, then $2\sqrt{3} = \sqrt{12}$

$(\sqrt{2} - 1)^2$ on the bottom

→ expand it first: $3 - 2\sqrt{2}$

“show each stage of your working”

→ conjugate line, expanded top, bottom $a^2 - b$

Change the sign

× top and bottom

Bottom = $a^2 - b$

Simplify

WORKED MODEL • JUNE 2025 P2HR

Show that $\frac{4}{3\sqrt{5} + 7}$ can be written as $a - \sqrt{b}$.

1
$$\frac{4(3\sqrt{5} - 7)}{(3\sqrt{5} + 7)(3\sqrt{5} - 7)}$$

conjugate

2
$$\frac{12\sqrt{5} - 28}{45 - 49} = \frac{12\sqrt{5} - 28}{-4}$$

bottom $a^2 - b$

3
$$7 - 3\sqrt{5}$$

divide

ANSWER $7 - \sqrt{45}$

× **TRAP** $(3\sqrt{5})^2 = 45$, not 15. Square the 3 as well.

IDEA 04

Practise: Rationalise: Two Terms

My questions first, then selected practice, then real past papers.

Q19 DR ESLAM Conjugate

CH 1 • TOPIC 10 3 marks

Rationalise $\frac{4}{3 - \sqrt{5}}$. Give your answer as $a + \sqrt{b}$.

SELECTED PRACTICE**Q20** PRACTICE Practice • Hard

CH 1 • TOPIC 10 3 marks

8 Rationalise the denominator of $\frac{4}{7 - \sqrt{5}}$

Show each stage of your working.

Give your answer in the form $a + b\sqrt{5}$ where a and b are fractions in their simplest forms.

(3 marks)

Q21 PRACTICE Practice • Hard

CH 1 • TOPIC 10 2 marks

12 Express $\frac{2}{\sqrt{3}-1}$ in the form $p + \sqrt{q}$

where p and q are integers.
Show your working clearly.

(2 marks)

Q22 PRACTICE Practice • Very Hard

CH 1 • TOPIC 10 3 marks

5 Rationalise the denominator of $\frac{a + \sqrt{4b}}{a - \sqrt{4b}}$ where a is an integer and b is a prime number.

Simplify your answer.

(3 marks)

Q23**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 10

4 marks

10 Show that $\frac{(4 + 2\sqrt{5})}{\sqrt{5} - 1}$ can be simplified to $\frac{3\sqrt{5} + 7}{2}$

(4 marks)

PAST PAPERS**Q24****PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q19

CH 1 • TOPIC 10

3 marks

19 Without using a calculator, rationalise the denominator of $\frac{6}{3 - \sqrt{7}}$

Simplify your answer.

You must show each stage of your working.

(Total for Question 19 is 3 marks)

Q25**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q17

CH 1 • TOPIC 10**3 marks**

17 Express $\frac{8}{\sqrt{5}-1}$ in the form $\sqrt{a} + b$ where a and b are integers.

Show each stage of your working clearly.

.....

(Total for Question 17 is 3 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q17

CH 1 • TOPIC 10**3 marks**

17 Show that $\frac{\sqrt{12}}{\sqrt{3}+2}$

can be written in the form $a + \sqrt{b}$ where a and b are integers.

(Total for Question 17 is 3 marks)

Q27**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q21

CH 1 • TOPIC 10**4 marks**

- 21 Express $\frac{3 + \sqrt{8}}{(\sqrt{2} - 1)^2}$ in the form $p + \sqrt{q}$ where p and q are integers.

Show each stage of your working clearly.

(Total for Question 21 is 4 marks)

Q28**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1H • Q16

CH 1 • TOPIC 10**3 marks**

- 16 Without using a calculator, show that $\frac{12}{\sqrt{2} - 1} - (\sqrt{2})^5 = 2\sqrt{32} + 12$

Show your working clearly.

(Total for Question 16 is 3 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q16

CH 1 • TOPIC 10**3 marks**

- 16 Show that $\frac{2\sqrt{3}}{\sqrt{3}-1}$ can be written in the form $a + \sqrt{a}$ where a is an integer.

Show your working clearly.

(Total for Question 16 is 3 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q17

CH 1 • TOPIC 10**3 marks**

- 17 Show that $\frac{1+\sqrt{5}}{3-\sqrt{5}}$ can be written in the form $a + \sqrt{b}$ where a and b are integers.

Show each stage of your working clearly.

(Total for Question 17 is 3 marks)

Q31**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q17

CH 1 • TOPIC 10**4 marks**

17 (a) Express $\sqrt{675}$ in the form $n\sqrt{27}$ where n is a positive integer.

(b) Show that $\frac{5 - \sqrt{2}}{\sqrt{2} - 1}$ can be written in the form $a + b\sqrt{2}$ where a and b are integers.

(Total for Question 17 is 4 marks)

Q32 PAST PAPER Edexcel 4MA1 • June 2025 • Paper 1H • Q15

CH 1 • TOPIC 10 4 marks

15 $(\sqrt{3})^5 = k\sqrt{3}$ where k is an integer.

(a) Find the value of k

$k = \dots\dots\dots$

(b) Show that $\frac{21}{3 - \sqrt{2}}$ can be written in the form $c + \sqrt{d}$

where c and d are integers.

Show each stage of your working clearly.

(Total for Question 15 is 4 marks)

Q33 PAST PAPER Edexcel 4MA1 • June 2025 • Paper 2HR • Q16

CH 1 • TOPIC 10 3 marks

16 Show that $\frac{4}{3\sqrt{5} + 7}$ can be written in the form $a - \sqrt{b}$ where a and b are integers.

Show each stage of your working.

(Total for Question 16 is 3 marks)

Q34 **PAST PAPER** Edexcel 4MA1 • November 2025 • Paper 1H • Q17**CH 1 • TOPIC 10** **3 marks**

- 17 Show that $\frac{2\sqrt{7} + 2}{\sqrt{7} - 3}$ can be written in the form $a - \sqrt{b}$ where a and b are integers.

Show your working clearly.

(Total for Question 17 is 3 marks)

05 Surds in Problems

Lengths are given exactly. Set up the equation as usual, then rationalise to find the unknown.

WHEN YOU SEE → YOU DO

“area 18, length $\sqrt{7} + 1$, find the width”

→ width = $\frac{18}{\sqrt{7} + 1}$, rationalise

volume of a cuboid with $x, x, 2\sqrt{5} - 3$

→ $x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3}$

“ x for which R is a maximum”

→ vertex of the quadratic, then rationalise

WORKED MODEL • JANUARY 2020 P2H

A rectangle has area 18 cm^2 and length $(\sqrt{7} + 1) \text{ cm}$. Find the width as $a\sqrt{b} + c$.

$$1 \quad \frac{18}{\sqrt{7} + 1} \times \frac{\sqrt{7} - 1}{\sqrt{7} - 1}$$

width = area ÷ length

$$2 \quad \frac{18(\sqrt{7} - 1)}{6}$$

bottom $7 - 1$

ANSWER $3\sqrt{7} - 3$

IDEA 05

Practise: Surds in Problems

My questions first, then real past papers.

PAST PAPERS**Q35****PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2H • Q20

CH 1 • TOPIC 10**3 marks**

20 The area of a rectangle is 18 cm^2

The length of the rectangle is $(\sqrt{7} + 1)\text{ cm}$.

Without using a calculator and showing each stage of your working,

find the width of the rectangle.

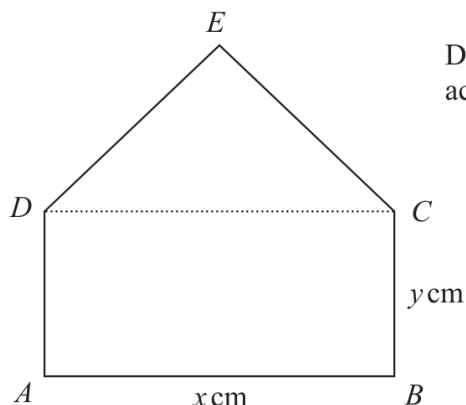
Give your answer in the form $a\sqrt{b} + c$ where a , b and c are integers.

..... cm

(Total for Question 20 is 3 marks)

Q36 PAST PAPER Edexcel 4MA1 • November 2021 • Paper 1H • Q19

CH 1 • TOPIC 10 6 marks

19 $ABCED$ is a five-sided shape.Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.
 CED is an equilateral triangle.

$$AB = x \text{ cm} \quad BC = y \text{ cm}$$

The perimeter of $ABCED$ is 100 cm.The area of $ABCED$ is $R \text{ cm}^2$

(a) Show that $R = \frac{x}{4} \left(200 - [6 - \sqrt{3}]x \right)$

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$x = \dots\dots\dots$

(ii) Explain why the maximum value of R is given by this value of x .

.....

(Total for Question 19 is 6 marks)

Q37 PAST PAPER Edexcel 4MA1 • June 2023 • Paper 2H • Q23

CH 1 • TOPIC 10 4 marks

23 The diagram shows a cuboid with a square cross section.

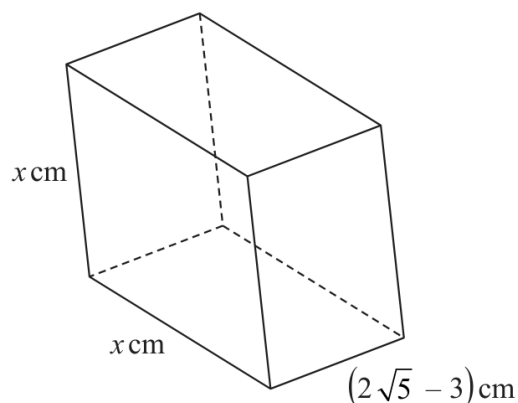


Diagram **NOT**
accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5}) \text{ cm}^3$

Without using a calculator, find the value of x

Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

Show your working clearly.

$x = \dots\dots\dots$

(Total for Question 23 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE 1

$$\sqrt{ab} = \sqrt{a}\sqrt{b}$$

square factor out

ZONE 2

$$\sqrt{a}\sqrt{a} = a$$

ZONE 3

$$(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$$

- | | | |
|-----------|------------------|---|
| 01 | Simplify | Biggest square factor out; only like surds add. |
| 02 | Expand | Like algebra; $\sqrt{a}\sqrt{a} = a$; match parts. |
| 03 | One term | Multiply top and bottom by the surd. |
| 04 | Two terms | Conjugate; bottom $a^2 - b$. |
| 05 | Problems | Set up, then rationalise. |

BOTTOM LINE Never a surd on the bottom → one term: times itself → two terms: change the sign.

Answers

Surds • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • SIMPLIFY A SURD

- Q1** Dr Eslam • Square factor (a) $5\sqrt{2}$ (b) $2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$
- Q2** Practice • Medium $\frac{10\sqrt{5}}{5} = 2\sqrt{5}$
- Q3** Practice • Medium $3\sqrt{5} + 2\sqrt{5}$
- Q4** Practice • Medium $\sqrt{4}\sqrt{5}$
- Q5** Practice • Hard $\sqrt{40} + \sqrt{90} = 2\sqrt{10} + 3\sqrt{10}: a = 5$
- Q6** Practice • Hard (a) $\sqrt{3} + 2\sqrt{3}$ (b) $\frac{1}{27\sqrt{3}} = \frac{\sqrt{3}}{81}$

IDEA 02 • EXPAND BRACKETS WITH SURDS

- Q7** Dr Eslam • Expand $4 + 4\sqrt{3} + 3 = 7 + 4\sqrt{3}$
- Q8** Practice • Medium (a) $\frac{12\sqrt{3}}{3}$ (b) $2 + 2\sqrt{16} + 8 = 18$
- Q9** Practice • Hard (a) $\frac{5\sqrt{2}}{2}$ (b) $(7 + 4\sqrt{3}) - (7 - 4\sqrt{3})$
- Q10** Practice • Hard $\frac{31}{\sqrt{31}} = \sqrt{31}$
- Q11** Practice • Very Hard $T = 16\sqrt{2}: c = 16$
- Q12** Edexcel 4MA1 • January 2021 • Paper 1H • Q18 $x = 19, y = 3$
- Q13** Edexcel 4MA1 • January 2022 • Paper 2HR • Q13 $a^2 - b^2 = 32\sqrt{2} = 8\sqrt{32}: y = 8$

IDEA 03 • RATIONALISE: ONE TERM

Q14 Dr Eslam • One term

$$\frac{6\sqrt{2}}{2} = 3\sqrt{2}$$

Q15 Practice • Medium

$$\frac{6\sqrt{5}}{\sqrt{3}} = \sqrt{60}$$

Q16 Practice • Hard

(a) $4 - 3 = 1$ on the bottom, not 7 (b) $\sqrt{12} \times 3$
 wrong: $\sqrt{12} = 2\sqrt{3}$, answer $\frac{5\sqrt{3}}{6}$
 $\sqrt{135}$

Q17 Practice • Hard

Q18 Practice • Very Hard

$$\frac{\sqrt{2}}{\sqrt{2} + 1} = \sqrt{2}(\sqrt{2} - 1) = 2 - \sqrt{2}$$

IDEA 04 • RATIONALISE: TWO TERMS

Q19	Dr Eslam • Conjugate	$\frac{4(3 + \sqrt{5})}{4} = 3 + \sqrt{5}$
Q20	Practice • Hard	$\frac{7}{11} + \frac{1}{11}\sqrt{5}$
Q21	Practice • Hard	$1 + \sqrt{3}$
Q22	Practice • Very Hard	$\frac{a^2 + 4a\sqrt{b} + 4b}{a^2 - 4b}$
Q23	Practice • Very Hard	$\frac{(4 + 2\sqrt{5})(\sqrt{5} + 1)}{4} = \frac{14 + 6\sqrt{5}}{4}$
Q24	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q19	$9 + 3\sqrt{7}$
Q25	Edexcel 4MA1 • November 2021 • Paper 2H • Q17	$\sqrt{20} + 2$
Q26	Edexcel 4MA1 • January 2022 • Paper 2H • Q17	$-6 + \sqrt{48}$
Q27	Edexcel 4MA1 • June 2022 • Paper 1HR • Q21	$17 + \sqrt{288}$
Q28	Edexcel 4MA1 • June 2022 • Paper 1H • Q16	$12\sqrt{2} + 12 - 4\sqrt{2} = 8\sqrt{2} + 12 = 2\sqrt{32} + 12$
Q29	Edexcel 4MA1 • June 2023 • Paper 2HR • Q16	$3 + \sqrt{3}$
Q30	Edexcel 4MA1 • May 2024 • Paper 1H • Q17	$2 + \sqrt{5}$
Q31	Edexcel 4MA1 • November 2024 • Paper 1H • Q17	(a) $n = 5$ (b) $3 + 4\sqrt{2}$
Q32	Edexcel 4MA1 • June 2025 • Paper 1H • Q15	(a) $k = 9$ (b) $9 + \sqrt{18}$
Q33	Edexcel 4MA1 • June 2025 • Paper 2HR • Q16	$7 - \sqrt{45}$
Q34	Edexcel 4MA1 • November 2025 • Paper 1H • Q17	$-10 - \sqrt{112}$

IDEA 05 • SURDS IN PROBLEMS

Q35Edexcel 4MA1 • January 2020 •
Paper 2H • Q20

$$3\sqrt{7} - 3$$

Q36Edexcel 4MA1 • November 2021
• Paper 1H • Q19

(b)(i) $x = \frac{100}{6 - \sqrt{3}}$ (ii) R is a quadratic in x with
negative x^2 : the vertex is the maximum
 $x^2 = 9 + 4\sqrt{5}$: $x = 2 + \sqrt{5}$

Q37Edexcel 4MA1 • June 2023 • Pa-
per 2H • Q23